MACHINE LEARNING

Time Series Forecasting with ARIMA , SARIMA and SARIM

A deep-dive on the gold standard of time series forec

Brendan Artley

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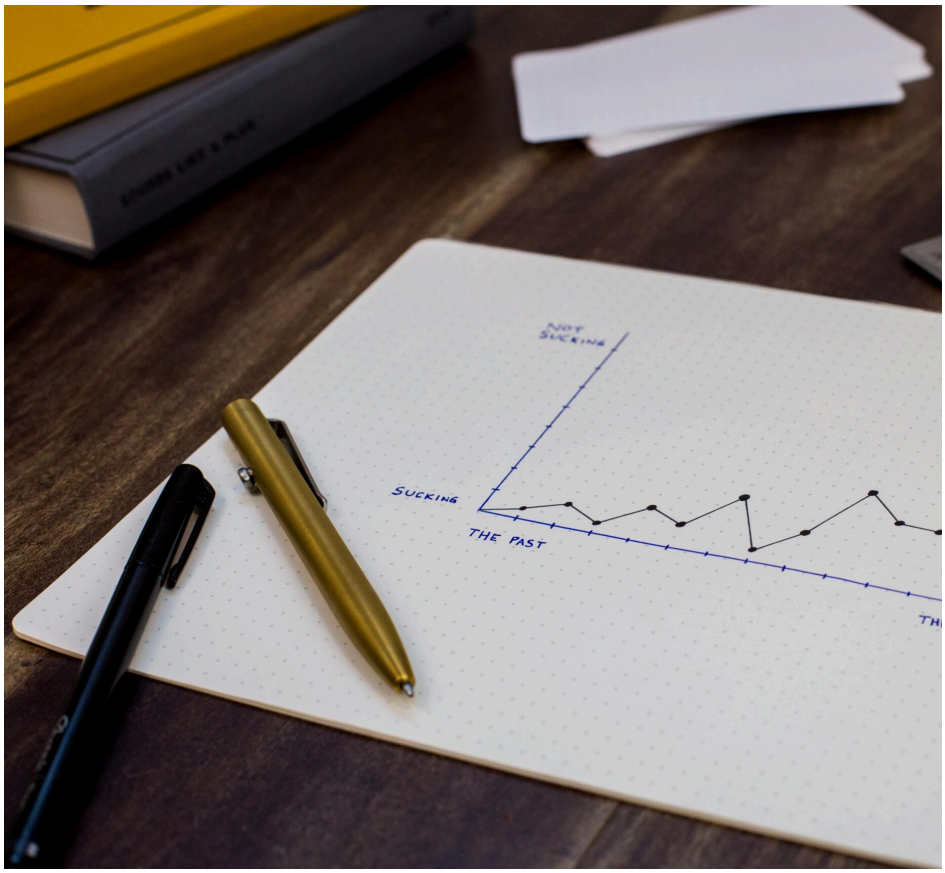


Time Series Graph – By Isaac Smith

Time series forecasting is a difficult problem with no easy answer. There are countless statistical models that claim to outperform each other, yet it is never clear which model is best.

That being said, ARMA-based models are often a good model to start with. They can achieve decent scores on most time-series problems and are well-suited as a baseline model in any time series problem.

This article is a comprehensive, beginner-friendl
you understand ARIMA-based models.

Introduction

The ARIMA model acronym stands for "Auto-Reg
Moving Average" and for this article we will will l
AR, I, and MA.

Autoregressive Component – AR(p)

The autoregressive component of the ARIMA mo
represented by AR(p), with the p parameter dete
number of lagged series that we use.

$$y_t = c + \sum_{n=1}^{p} \alpha_n y_{t-n} + \epsilon_t$$

AR Formula – By Author

AR(0): White Noise

If we set the p parameter as zero (AR(0)), with no autoregressive terms. This time series is just white noise. Each data point is sampled from a distribution with a mean of 0 and a variance of sigma-squared. This results in a sequence of random numbers that can't be predicted. This is really useful as it can serve as a null hypothesis, and protect our analyses from accepting false-positive patterns.

AR(1): Random Walks and Oscillations

With the p parameter set to 1, we are taking int
previous timestamp adjusted by a multiplier, an
white noise. If the multiplier is 0 then we get wh
the multiplier is 1 we get a random walk. If the r
between $0 < \alpha_1 < 1$, then the time series will exh
reversion. This means that the values tend to ho
revert to the mean after regressing from it.

AR(p): Higher-order terms

Increasing the p parameter even further is just r
further back and adding more timestamps adjus
multipliers. We can go as far back as we want, b
further back it is more likely that we should use
parameters such as the moving average (MA(q)).

Moving Average – MA(q)

"This component is not a rolling average, but rat
the white noise." – Matt Sosna

MA(q)

MA(q) is the moving average model and q is the number of lagged forecasting error terms in the prediction. In an MA(1) model, our forecast is a constant term plus the previous white noise term times a multiplier, added with the current white noise term. This is just simple probability + statistics, as we are adjusting our forecast based on previous white noise terms.

ARMA and ARIMA Models

ARMA and ARIMA architectures are just the AR (
and MA (Moving Average) components put toget

ARMA

The ARMA model is a constant plus the sum of
multipliers, plus the sum of the MA lags and the
white noise. This equation is the basis of all the
come next and is a framework for many forecas
across different domains.

ARIMA

$$Y_t = \beta_2 + \omega_1 \varepsilon_{t-1} + \omega_2 \varepsilon_{t-2} + \ldots +$$

ARIMA Formula – By Author

The ARIMA model is an ARMA model yet with a
step included in the model that we represent u
ifference order, which is the number of transfo

to make the data stationary. So, an ARIMA model is simply an ARMA model on the differenced time series.

SARIMA, ARIMAX, SARIMAX Models

The ARIMA model is great, but to include seasonality and exogenous variables in the model can be extremely powerful. Since the ARIMA model assumes that the time series is stationary, we need to use a different model.

SARIMA

$$y_t = c + \sum_{n=1}^{p} \alpha_n y_{t-n} + \sum_{n=1}^{q} \theta_n \epsilon_{t-n} + \sum_{n=1}^{P} \phi_n y_{t-sn} +$$

SARIMA Formula – By Author

Enter SARIMA (Seasonal ARIMA). This model is v
ARIMA model, except that there is an additional
autoregressive and moving average components
lags are offset by the frequency of seasonality (
24 – hourly).

SARIMA models allow for differencing data by se
yet also by non-seasonal differencing. Knowing
are best can be made easier through automatic
frameworks such as pmdarina.

ARIMAX and SARIMAX

$$d_t = c + \sum_{n=1}^{p} \alpha_n d_{t-n} + \sum_{n=1}^{q} \theta_n \epsilon_{t-n} + \sum_{n=1}^{r} \beta_n x_{n_t} + \sum_{n=1}^{P} \phi_n d_{t-sn} + \sum_{n=1}^{Q} \eta_n \epsilon_{t-sn} + \epsilon_t$$

Sarimax Formula – By Author

Above is the the of the SARIMAX model. This model takes into account exogenous variables, or in other words, use external data in our forecast. Some real-world examples of exogenous variables include gold price, oil price, outdoor temperatur

It is interesting to think that all exogenous facto technically indirectly modeled in the historical r That being said, if we include external data, the respond much quicker to its affect than if we re influence of lagging terms.

Code Example

Lets look at these models in actions through a s example in Python.

Loading Data

For this example, we are going to use the Air Pa This dataset contains the number of air travel p the start of 1949 to the end of 1960.

This dataset has a positive trend and annual sea

As soon as the dataset is read, the index is set t is standard practice when working with time-se

Pandas, and makes it easier to implement ARIMA, SARIMA, and SARIMAX.

	#Passengers
Month	
1949-01-01	112
1949-02-01	118
1949-03-01	132
1949-04-01	129
1949-05-01	121

Cell output – By Author

Trend

The general direction of the data over time. For
looking at the height of a newborn baby, their he
an upward trend into their youth. On the other h
a successful weight loss program will see their v
downward trend over time.

Seasonality + Cycles

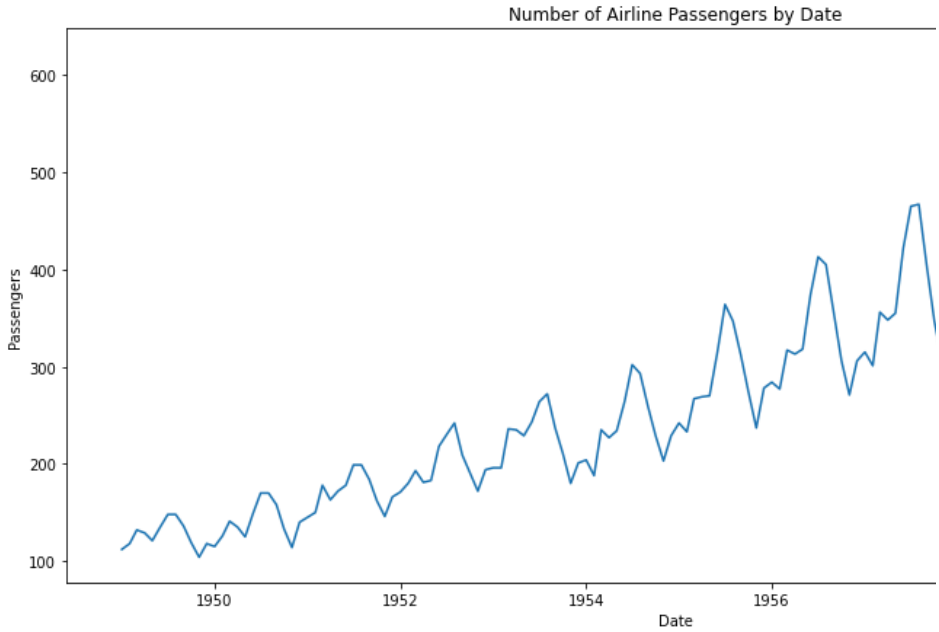
Any seasonal or repeating patterns with a fixed
be hourly, monthly, daily, annually, etc. One exar
that Winter Jacket sales increase in the winter r
decrease in the summer months. Another exam
be the balance of your bank account. In the 10 c

of every month, your balance follows a downward trend as you pay monthly rent, utilities, and other bill payments.

Irregularities + Noise

This is any large spikes or troughs in the data. One example of this could be your heart rate when you run the 400-meter dash. When you start the race your heart rate is similar to what it has been throughout the day, but during the race, it higher level for a small period of time before ret normal level.

In the visualization of the airline passenger data look for these components. At first glance, there positive trend and some sort of seasonality or c dataset. There does not appear to be any major noise in the data.

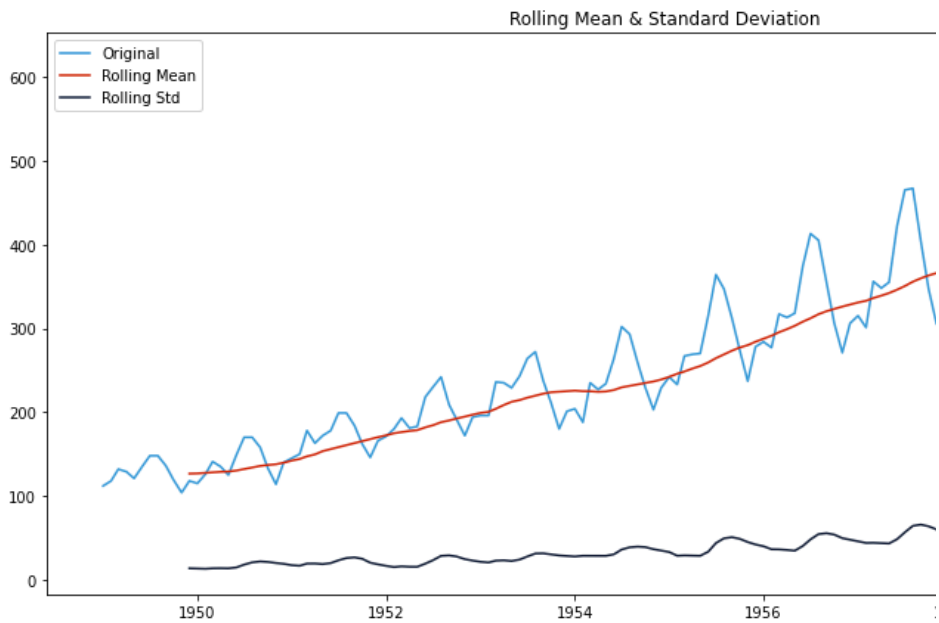


Airline Passengers Line Plot – By Author

Rolling Statistics

A rolling average is a great way to visualize how the dataset is trending. As the dataset provides counts by month, a window size of 12 will give us the annual rolling average.

We will also include the rolling standard deviation to see how much the data varies from the rolling average.



Rolling Statistics Plot – By Author

Augmented Dickey–Fuller Test

The Augmented Dickey-Fuller Test is used to de
series data is stationary or not. Similar to a t-te
significance level before the test and make con
hypothesis based on the resulting p-value.

Null Hypothesis: The data is not stationary.

lternative Hypothesis: The data is stationary.

For the data to be stationary (ie. reject the null hypothesis), the ADF test should have:

- p-value <= significance level (0.01, 0.05, 0.10, etc.)

If the p-value is greater than the significance level then we can say that it is likely that the data is not stationary.

We can see in the ADF test below that the p-val
meaning that it is very likely that the data is not

```
Results of Dickey Fuller Test:
Test Statistic
p-value
#Lags Used
Number of Observations Used
Critical Value (1%)
Critical Value (5%)
Critical Value (10%)
dtype: float64
```

ADF Test Output – By Author

ARIMA Model Selection w/ Auto-ARIMA

Although our data is almost certainly not statior
0.991), let's see how well a standard ARIMA mod
the time series

Using the auto_arima() function from the `pmdarima` package, we can perform a parameter search for the optimal values of the model.

Model Diagnostics

Four plots result from the plot_diagnostics function. The Standardized residual, Histogram plus KDE estir
and the correlogram.

We can interpret the model as a good fit based
conditions.

Standardized residual

There are no obvious patterns in the residuals, v
a mean of zero and having a uniform variance.

Histogram plus KDE estimate

The KDE curve should be very similar to the nor
(labeled as N(0,1) in the plot)

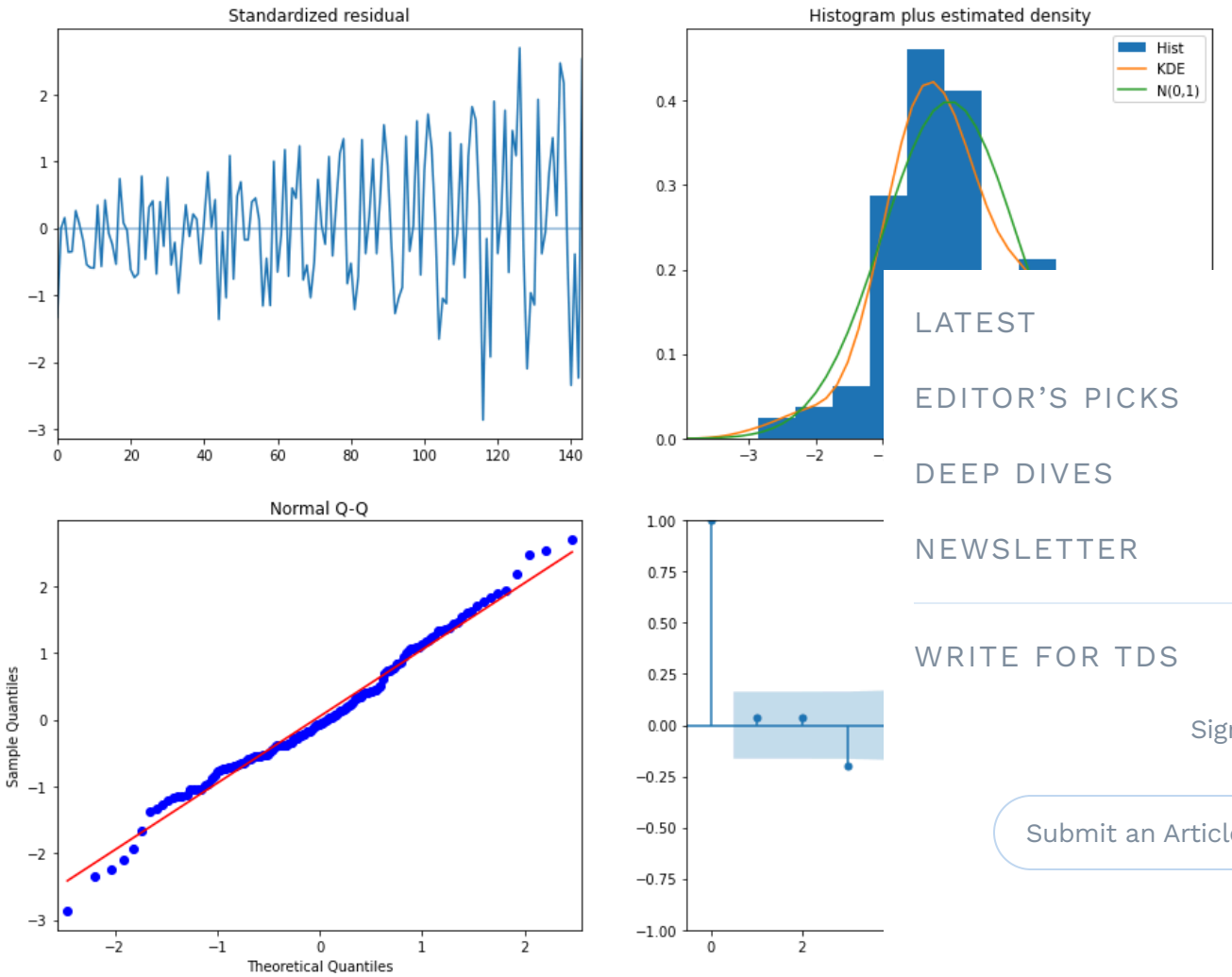
Normal Q-Q

Most of the data points should lie on the straigh

Correlogram (ACF plot)

95% of correlations for lag greater than zero sho
significant. The grey area is the confidence banc
utside of this then they are statistically signific

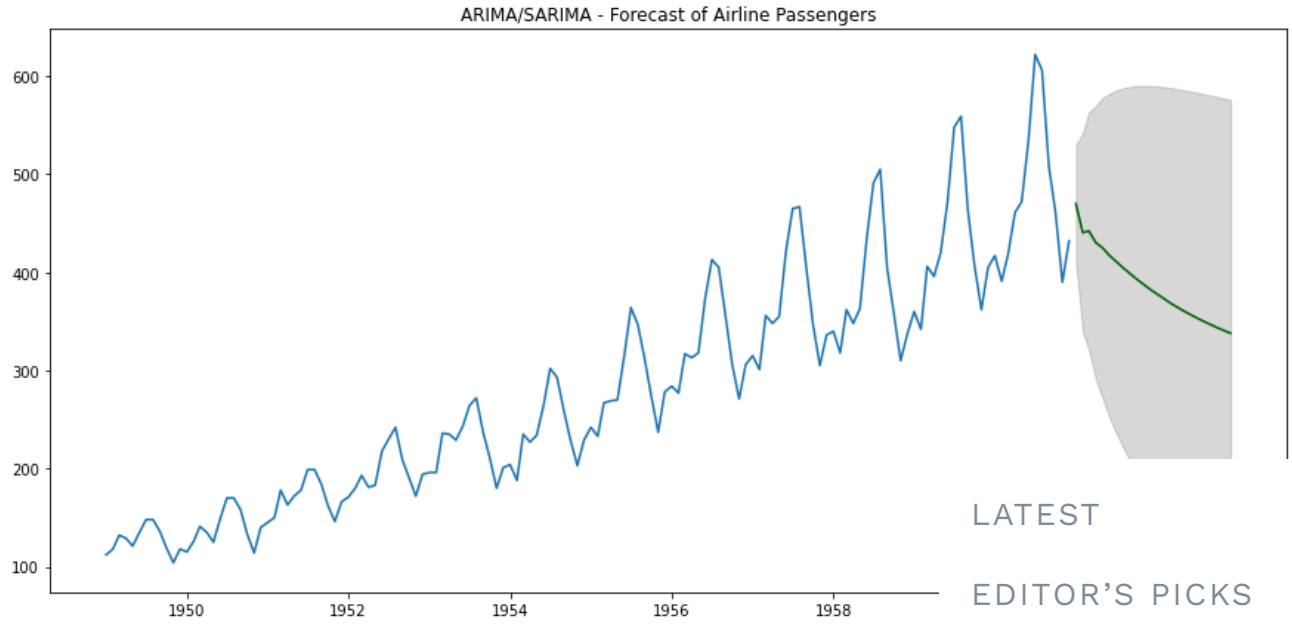
there are a few values outside of this area, and therefore we may need to add more predictors to make the model more accurate



Arima Diagnostics – By Author

We can then use the model to forecast airline p
over the next 24 months.

As we can see from the plot below, this doesn't
very accurate forecast. Maybe we need to chang
structure so that it takes into account seasonal



ARIMA Forecast – By Author

SARIMA Model

Now let's try the same strategy as we did above
a SARIMA model so that we can account for sea

Taking a look at the model diagnostics, we can s
significant differences when compared with the
model.

Standardized residual

The Standardized residual is much more consist
graph, meaning that the data is closer to being s

Histogram plus KDE estimate

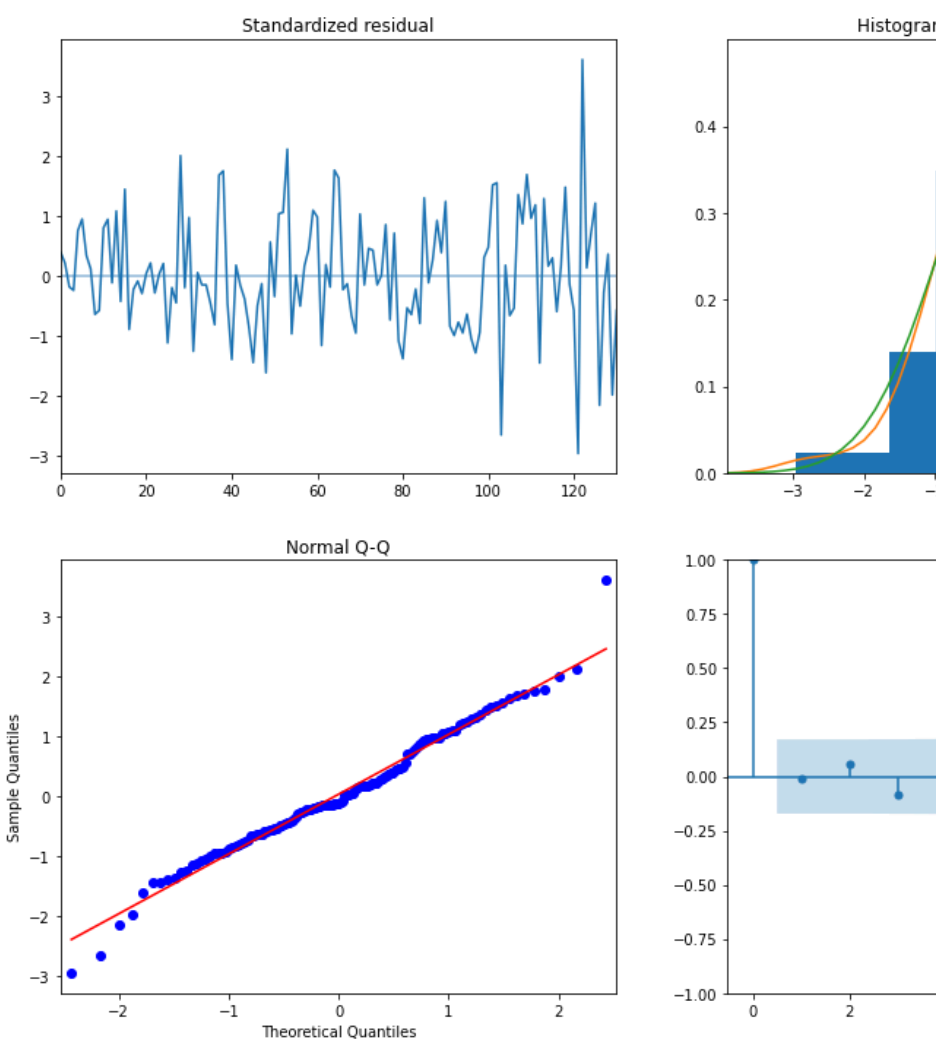
The KDE curve is similar to the normal distribut
changed here).

Normal Q-Q

The data points are clustered much closer to the line than in the ARIMA diagnostic plot.

Correlogram (ACF plot)

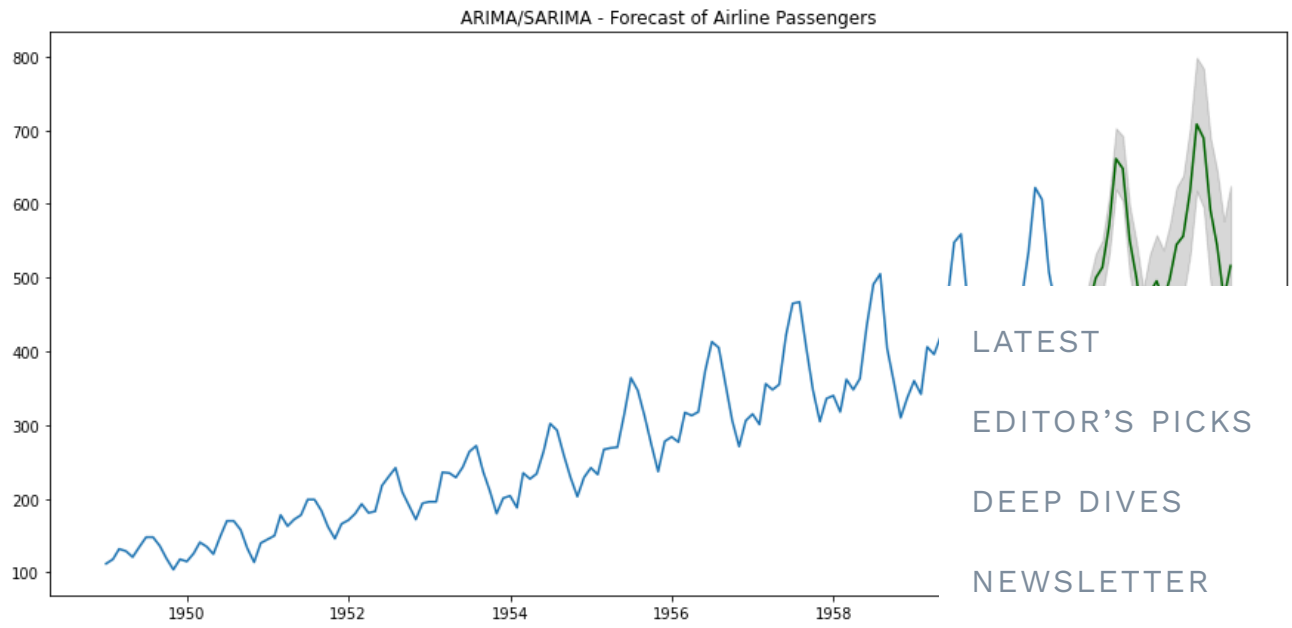
The grey area is the confidence band, and if values fall outside of this then they are statistically significant. We want all values inside this area. Adding the seasonality component did this! All the points now fall within the 95% confidence ir



Sarima Diagnostics

We can then use the model to forecast airline p
over the next 24 months as we did before.

As we can see from the plot below, this seems to be much more accurate than the standard ARIMA model!



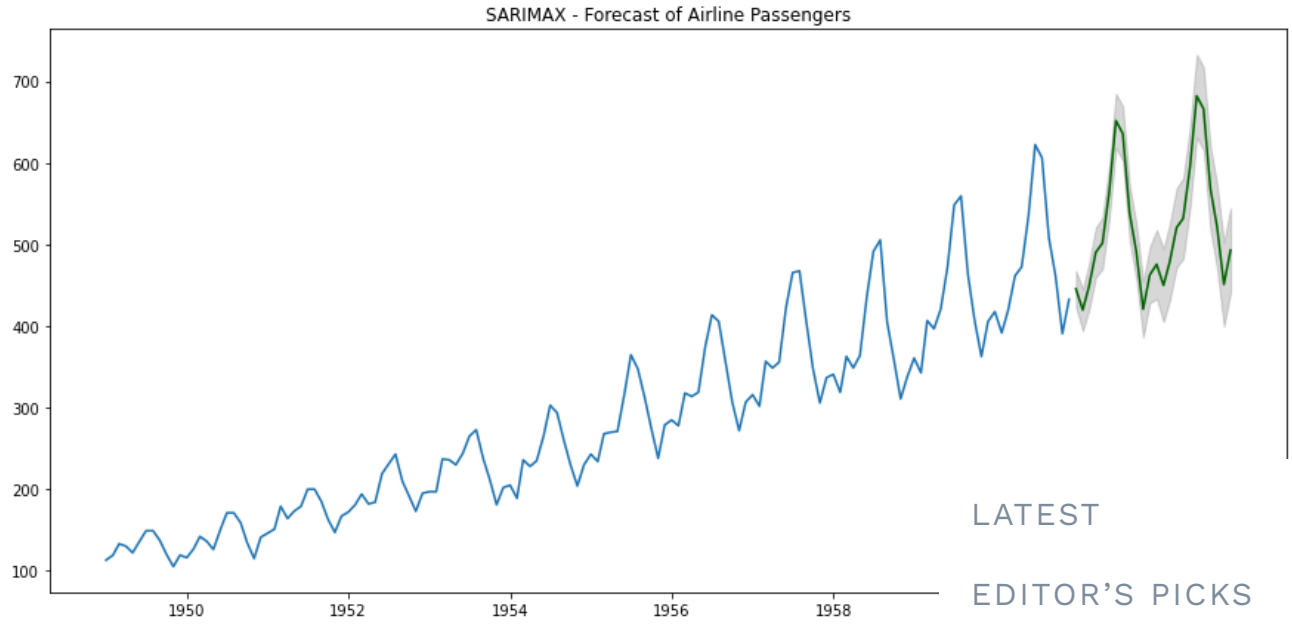
SARIMA Forecast – By Author

SARIMAX Model Selection

Now let's practice adding in an exogenous variab
example, I am simply going to add the month nu
exogenous variable, but this is not super useful
conveyed through the seasonality.

Note that we are adding additional square brack
data being passed into the SARIMAX model.

We can see from the following predictions that
some pretty good-looking predictions and the w
forecasted confidence interval has decreased. T
the model is more certain of its predictions.



SARIMAX Forecast – By Author

Closing Thoughts

Please find the code for this article here.

Putting ideas into my own words and implemen
models hands-on is the best way to learn. Hope
can motivate others to do the same.

ARIMA model architectures provide more explai
RNN's, yet RNN's are known to generate more a
predictions. Now I have a good grasp on the ARI
architecture, I need to look into LSTM and RNN
models for forecasting time series data!

Further Reading

Throughout the notebook I implement and rewo
following sources. Thank you to all for sharing!

A Deep Dive on Arima Models – by Matt Sosna ←

Time Series For beginners with ARIMA — by @Arindam Chatterjee

Arima Model for Time Series Forecasting – by @Prashant Banerjee

StatsModels ADF Documentation

Removing Trends and Seasonality Article — by Jason Brownlee

A Gentle Introduction to SARIMA — by Jason Br

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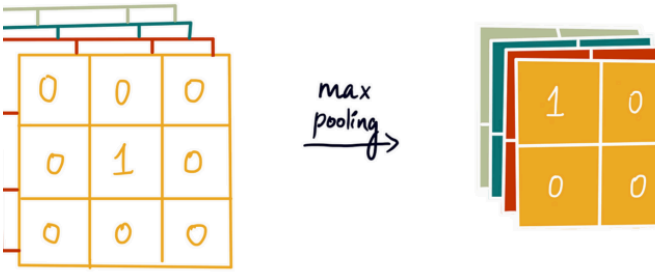

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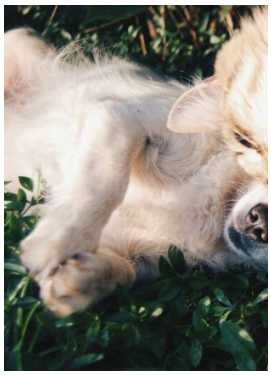

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A beginner's guid reconciliation

Dr. Robert Kübler

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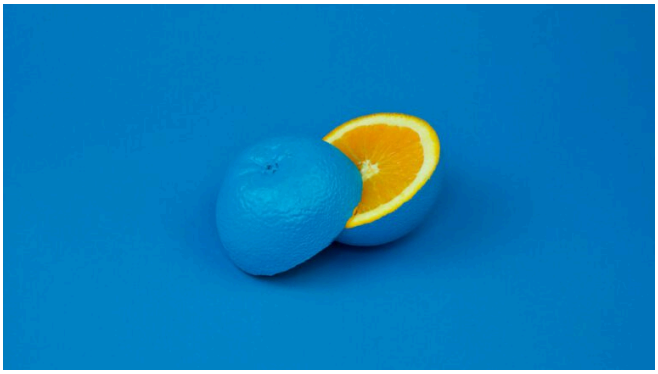

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